

Instructions for the Examinations Officer

Pearson Edexcel International A level Mathematics Mechanics Paper 1 WME01
Date: 10 January 2025

There is an error on page 2 of this question paper. A Clarification Notice is enclosed below.

At the start of the examination the Clarification Notice must be read aloud to the candidates taking the examination.

All other questions on the paper have been printed correctly.

Clarification Notice

There is an error on page 2, Question 1b.

Please amend your question paper as follows:

The question currently reads:

- The resultant of the four forces acts in **the direction of** the vector $(7\mathbf{i} + 2\mathbf{j})$.

It should read:

- The resultant of the four forces acts in **a direction that is parallel to** the vector $(7\mathbf{i} + 2\mathbf{j})$.

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel International Advanced Level

Friday 10 January 2025

Afternoon (Time: 1 hour 30 minutes)

Paper reference **WME01/01**

Mathematics

International Advanced Subsidiary/Advanced Level

Mechanics M1

You must have:
Mathematical Formulae and Statistical Tables (Yellow), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$, and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Turn over ►

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1. A particle of mass 2.5 kg moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (6\mathbf{i} + 8\mathbf{j})\text{ N}$$

$$\mathbf{F}_2 = (-16\mathbf{i} + 2\mathbf{j})\text{ N}$$

$$\mathbf{F}_3 = (-2\mathbf{i} + 8\mathbf{j})\text{ N}$$

- (a) Find the magnitude of the acceleration of the particle.

(4)

A fourth force, $\mathbf{F}_4 = (p\mathbf{i} + p\mathbf{j})\text{ N}$, where p is a constant, is added.

The resultant of the four forces acts in a direction that is parallel to the vector $(7\mathbf{i} + 2\mathbf{j})$.

- (b) Find the value of p .

(4)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 8 marks)



2. The fixed points A , B and C lie in a straight line on a horizontal road.

- At time $t = 0$, a motorbike passes through A with speed 5 m s^{-1}
- From A , the motorbike accelerates uniformly until it reaches B with a speed of $V \text{ m s}^{-1}$
- The motorbike takes T_1 seconds to travel from A to B
- From B , the motorbike decelerates uniformly until it comes to rest at C
- The motorbike takes T_2 seconds to travel from B to C

(a) Sketch a speed-time graph for the motion of the motorbike as it moves from A to C .

(3)

The distance AB is 132 m and the distance BC is 136 m.

(b) Find, in terms of V , an expression for

(i) T_1

(ii) T_2

(4)

Given that the motorbike takes 28 s to travel from A to C ,

(c) find the value of V ,

(2)

(d) find the deceleration of the motorbike.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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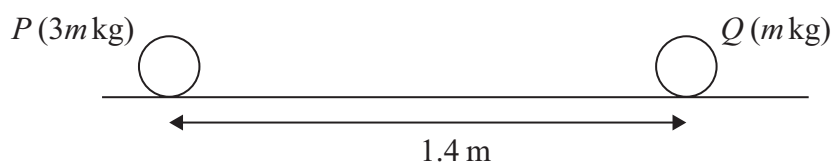
Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 11 marks)



3.

**Figure 1**

A particle P of mass $3m \text{ kg}$ and a particle Q of mass $m \text{ kg}$ are at rest on a rough horizontal surface. The distance between P and Q is 1.4 m , as shown in Figure 1.

An impulse of magnitude $\lambda \text{ N s}$ is now applied to P in the direction PQ .
Immediately after the impulse is applied, the speed of P is 5 m s^{-1}

- (a) Find λ in terms of m . (2)

Immediately before P collides with Q , the speed of P is 2.5 m s^{-1}
The coefficient of friction between P and the surface is μ

- (b) Find the value of μ (7)

Immediately after P collides with Q , the speed of Q is 2.1 m s^{-1}

- (c) Find the speed of P immediately after the collision. (3)



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Question 3 continued

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Question 3 continued

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Question 3 continued

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(Total for Question 3 is 12 marks)



4.

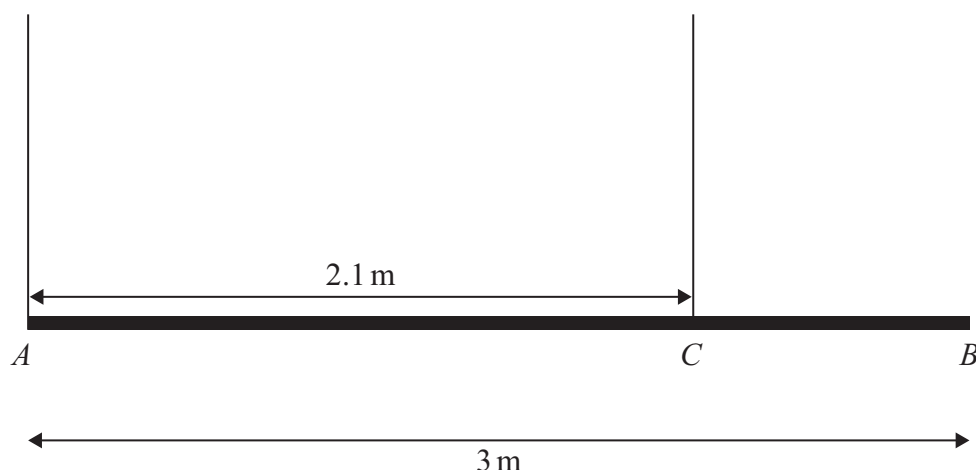


Figure 2

A uniform rod AB has length 3 m and weight W newtons.

The rod is suspended by two light vertical ropes.

One rope is attached to the rod at A and the other rope is attached to the rod at C , where $AC = 2.1$ m.

The rod is in equilibrium in a horizontal position, as shown in Figure 2.

The tension in the rope at C is 350 N.

(a) Show that $W = 490$

(3)

A particle P of weight 210 N is attached to the rod at a distance d metres from A .

The tension in the rope at C is now 600 N.

The rod remains in equilibrium in a horizontal position.

(b) Find the value of d .

(3)

Particle P is removed from the rod.

A particle Q of weight X newtons is now attached at B .

The rod remains in equilibrium in a horizontal position and is now on the point of tilting.

(c) Find the value of X .

(4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 10 marks)



5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors and position vectors are given relative to a fixed origin.]

In a game, a ball B is rolled across a horizontal surface towards a fixed target. The ball is modelled as a particle moving with constant velocity.

At time $t = 1$ s, the position vector of B is $(-2\mathbf{i} + 5\mathbf{j})$ m.

At time $t = 9$ s, the position vector of B is $(10\mathbf{i} - 3\mathbf{j})$ m.

- (a) Find the velocity of the ball.

(3)

The position vector of the target is $(13\mathbf{i} - 5\mathbf{j})\text{ m}$.

- (b) Use the model to find the distance of B from the target at time $t = 7$ s.

(4)

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Question 5 continued

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(Total for Question 5 is 7 marks)



6.

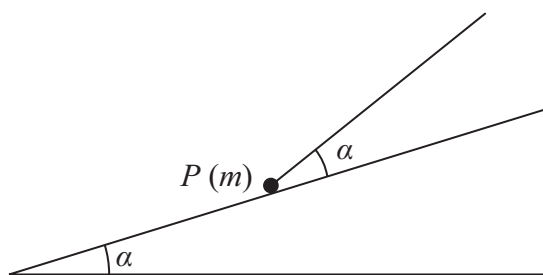


Figure 3

A particle P of mass m is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle α to the horizontal, where $\alpha < 45^\circ$

The string is inclined to the plane at angle α , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

When the tension in the string is 0.75 mg , P is on the point of moving up the plane.

- (a) Find an expression for the magnitude of the frictional force acting on P , giving your answer in terms of m , g and α .

(3)

The coefficient of friction between P and the plane is $\frac{1}{2}$.

- (b) Show that

$$\tan \alpha = \frac{2}{5}$$

(6)

The string breaks.

- (c) Determine whether P remains at rest. You must justify your reasoning.

(3)



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Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7.

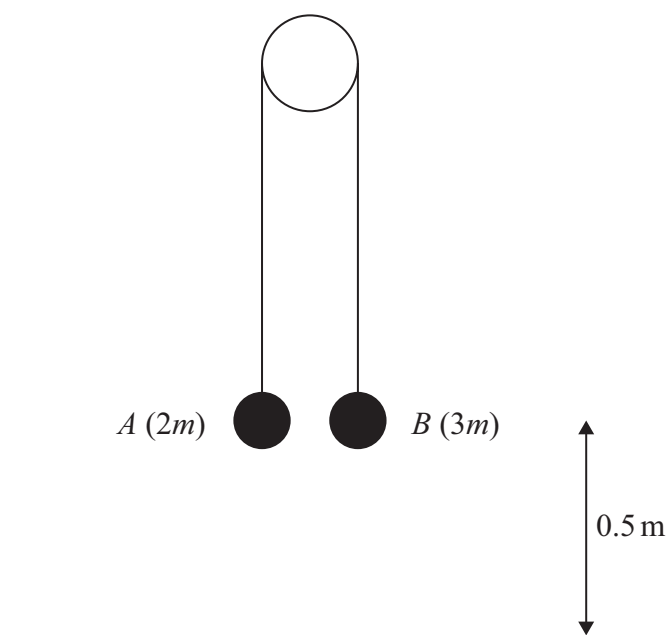


Figure 4

One end of a light inextensible string is attached to a particle A of mass $2m$.
 The other end is attached to a particle B of mass $3m$.
 The string passes over a small smooth fixed pulley.
 The string is taut and both straight parts of the string are vertical.
 Both particles are held at a distance 0.5 m above a horizontal surface, as shown in Figure 4.

The system is released from rest and B moves downwards.

(a) Find the tension in the string in terms of m and g . (5)

(b) Find the speed of B at the instant it strikes the surface. (4)

In the subsequent motion, A does not reach the pulley and B does not rebound after it strikes the surface.

(c) Find the time from the instant when the system is released from rest to the instant when A first reaches a height of 1.06 m above the surface. (6)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS

